

3. Neurons and Neural Communication

§3.1 — Computational: Why Neurons Evolved

§3.1.1 — The Computational Question

Module 1 adopted three levels of explanation, and it put the computational level first: what problem does this system solve, and why does solving it matter? Module 2 then spent a chapter on how we set systems beside each other and reason from what we find. This chapter is the first place in the course where we take a single thing and ask all three of Marr's questions about it, one section at a time, and so it is the first place where the computational question gets a section of its own. It is worth being exact about what that question asks. It is not a question about what a neuron is built out of; §3.3 asks that. It is not a question about the steps a neuron goes through; §3.2 asks that. It is a question about a problem, and about who had the problem.

For an organism, Module 1 welded the computational question to evolution, and that weld is what makes this section possible. A designed object has a purpose because somebody had a purpose for it. Living things have no designer. So when we ask what a biological structure is for, the only answer available that does not appeal to magic is a historical one: the structure is here because the organisms that had it left more descendants than the organisms that did not. Asking what a neuron is for is therefore asking what selective pressure favored building one.

That gives us the question this section answers. What could a neuron do that earlier life could not do, and for whom did that difference matter enough to be selected? The disciplines that answer questions in this shape are evolutionary biology and ecology. Each of the three sections of this chapter names its level and its discipline as it opens, because the rhythm is new and worth feeling as it starts.

Module 2 built natural selection from the ground up, and we will use it here rather than rebuild it. Variation arises among individuals. Some of that variation is heritable. Some heritable variants leave more descendants than others, and over generations the population shifts. Module 2 aimed that machinery at species and at traits. We are about to aim it somewhere narrower and stranger: at a kind of cell. A neuron is an adaptation in exactly the sense Module 2 gave the word, and the same three-part logic that explains the coloring of a moth explains why some lineages came to build neurons and most never did.

One correction from Module 2 has to travel with the logic, because this chapter is unusually good at inviting the mistake it corrects. Module 2 was firm that evolution is not a climb toward anything, and that there is no ladder with human beings at the top of it. This section walks forward through time, from chemistry to cells to animals, and a walk forward through time reads very easily as a walk upward. It is not one. Neurons are not an achievement that some lineages managed and others failed at. They are an expensive solution to a problem that only some ways of making a living actually have. Plants have no neurons and are not thereby behind. By every measure that matters to a plant, plants are doing extraordinarily well.

§3.1.2 — Before a Cell Can Afford Anything

A neuron is a cell, and an unusually elaborate one. Before we can ask how life came to afford such a thing, there has to be a cell at all, and a cell is not an obvious object. It is an arrangement of chemistry that holds itself together, repairs itself, powers itself, and makes more of itself. Nothing about ordinary chemistry suggests that such an arrangement should exist. So the story starts earlier

than an introduction to the mind usually starts: on an Earth with no life on it and a great deal of chemistry.

The leading account of what happened next begins with RNA. On the early Earth, unremarkable chemical reactions produced short chains of RNA, most of which did nothing at all. A few folded into shapes that could speed up chemical reactions, including the reaction that copies RNA. A molecule that speeds up a reaction is a catalyst, and an RNA molecule that does it is a *ribozyme*. Notice what a ribozyme that copies itself already has. The copying is imperfect, so there is variation. Copies resemble their originals, so there is inheritance. Some variants copy faster or last longer, so some leave more descendants than others. Those are Module 2's three ingredients, present in molecules, before anything exists that anyone would call an organism. Evolution starts before life does.

The rest of the cell arrives as a series of additions to that beginning. Fatty molecules form sheets that curl closed into bubbles, which holds a set of reactions together and keeps the ocean out. Proteins turn out to be far more versatile than RNA at catalyzing, moving, and sensing, so RNA's catalytic work is gradually handed to them by way of the ribosome, the machine that reads a sequence and builds a protein from it. DNA arrives last, as a chemically stabler archive than RNA for the sequences worth keeping. Every living thing yet discovered descends from a single population that already had all of this, called the last universal common ancestor, or *LUCA*.

None of that is a story we are free to tell however we like. Miller and Urey showed in the 1950s that simple gases and an electrical spark yield amino acids without any help, which is Module 2's synthetic method applied to the origin of life: build the conditions and see what they produce. The ribosome in every cell alive today still does its central chemical step with RNA rather than protein, which is close to a living fossil of the world that preceded proteins, and *LUCA* itself is a reconstruction, worked out by comparing the genomes of living organisms and asking what their common ancestor must have had.

One feature of this early world matters more for our purposes than any of the chemistry. Life was sensing its surroundings and acting on what it sensed long before it had neurons. A swimming bacterium can detect which way the concentration of a nutrient is increasing and steer up the gradient, a behavior called **chemotaxis**, and it can steer away from a substance that damages it in the same way. So the loop of taking something in from the world, doing something with it, and acting on the result is older than neurons by billions of years. A single cell already runs that loop, slowly, using proteins as its sensors and its motors. Whatever neurons added, they did not add behavior. It is worth noticing in passing that the cell was already processing information in another sense too, since DNA is read into RNA and RNA into protein, so the neuron will turn out to be a new layer of information processing rather than the first one.

§3.1.3 — The Energy Revolution

Everything a cell does costs something. Building a protein costs, moving a flagellum costs, repairing damage costs, and copying a genome costs a great deal. Cells pay all of these bills in the same currency: **ATP**, a small molecule that stores energy in a chemical bond and releases it where it is needed. A cell's whole economy runs on making ATP and spending it.

Early cells made ATP in a way that is ingenious and meager at once. Energy from chemical reactions is used to pump protons across a membrane, so that protons pile up on one side. They then flow back through a protein that spins as they pass, and the spinning drives the assembly of ATP. Water behind a dam is the usual comparison, and it is a good one, because the arrangement stores energy as a difference across a barrier and collects it on the way down. The scheme works, and it works in essentially every cell alive today. What it did not do, in early life, was produce very much. This is worth pausing on, because the natural assumption about early life is that it was simple and therefore easy. It was not easy. It was poor.

Then some microbes learned to capture sunlight. **Photosynthesis** uses light to build sugar out of carbon dioxide and water, which turns a renewable and enormous energy source into a storable

chemical one. The consequences for life were large. The consequences for the planet were larger, because photosynthesis has a waste product, and the waste product is oxygen.

Oxygen was at first a catastrophe. It is chemically aggressive, it wrecks the molecules of organisms not built to handle it, and its accumulation is associated with one of the earliest mass extinctions in Earth's history, known as the **Great Oxygenation Event**. What followed is the turn the rest of this chapter depends on. The same chemical aggressiveness that made oxygen a poison makes it an unusually good acceptor of electrons, and a lineage that could handle it could use it to extract far more energy from the same food. **Aerobic respiration** yields something like fifteen times as much ATP from a molecule of sugar as the older, oxygen-free methods do. Oxygen was a poison and then a windfall, and everything expensive that life built afterward was paid for out of that windfall.

§3.1.4 — What the Surplus Bought

A large and reliable energy surplus makes arrangements affordable that were not affordable before, and the history of life after oxygen is largely a history of newly affordable expenses.

The first is the cell itself getting bigger and more elaborate. A **eukaryote** is a cell with its DNA enclosed in a nucleus and its chemistry divided among internal compartments, and it is the kind of cell that plants, fungi, and animals are built from. Eukaryotes appear to have arisen through *endosymbiosis*: one cell engulfed another and, instead of digesting it, kept it. The engulfed oxygen-users became **mitochondria**, compartments dedicated to aerobic respiration, and a cell with its own internal power plants can pay for a great deal more machinery than one without them.

The second expense is bodies. **Multicellularity** is the arrangement in which cells that could live independently instead stay together and function as a single organism, and it has arisen many separate times. What it buys is *division of labor*. If every cell must do every job, every cell must be mediocre at all of them. If cells can specialize, some can be built for movement, others for digestion, others for sensing, and each can be far better at its one job than any generalist could be at any job. Tissues and organs are what specialization looks like once it has run for a while.

Specialization also creates a problem that did not exist before, and it is the problem this chapter is really about. A colony of identical cells has nothing much to coordinate. A body of specialized cells has to act as one thing. The cells that sense are not the cells that move, and the cells that move are not the cells that digest, so information has to get from the part of the body that has it to the part of the body that needs it, in time for it to be useful. Call this the coordination problem. Every complex multicellular organism has some version of it, and how urgently it needs solving turns out to depend entirely on how the organism makes a living.

§3.1.5 — Three Ways to Be Complex

Complex multicellular life did not solve the coordination problem once. It solved it at least three times, differently, and setting the three side by side isolates exactly the variable we are looking for (Table 3.1).

Reading across the table is Module 2's comparative method, applied not to two species but to three whole ways of making a living. Module 2's more common move asks why two things are alike. The useful move here is the mirror of it: these three are all complex, all multicellular, all descended from the same eukaryotic beginning, and only one of them evolved neurons. What separates that one from the other two is the thing we are looking for.

The separating variable is speed, and the reason is that animals eat other organisms and are eaten by them. A plant threatened by drought has hours or days to respond, and it responds with growth and chemistry, which is enough. An animal being hunted has a fraction of a second. In a predator-prey arms race, the animal that senses, decides, and acts slightly faster than its opponent is the animal that eats or the animal that escapes, and both outcomes get written into the next

generation. So the animal way of life puts a premium on speed that the other two strategies simply do not have.

Table 3.1 — *Three strategies for complex multicellular life, and what each one demands of a body. Only one of the three makes fast, whole-body coordination worth paying for.*

	Plants	Fungi	Animals
How it gets energy	Makes its own food from sunlight	Grows into its food and absorbs it	Finds food, and eats other organisms
What the body does	Stays put; grows toward light and water	Stays put; grows through the material it feeds on	Moves through the world, and is moved through by others
What can go wrong in a second	Very little	Very little	Being eaten; losing prey; a bad step
Coordination it needs	Slow and chemical is enough	Slow and chemical is enough	Fast, aimed, whole-body

That premium is worth naming, because it is the thing the rest of this chapter is about and the thing much of the rest of the course is about. Call it the **sense → decide → act loop**: take in information about the world, do something with it, and produce an action, fast enough for the action to still be the right one. It returns at the algorithmic level, where we ask what the neuron computes, and at the implementational level, where we ask what it costs to run. It is also not a problem unique to living things. Any system that has to operate on its own in a changing world faces some version of it, and a robot or a self-driving car in a later module needs its own answer to it. The neuron is evolution's answer.

The animal solution shows up early in development, and in one piece. Very early in the development of an animal embryo, at gastrulation, the cell layers that will become gut, muscle, and nervous system are laid down together, which is what we should expect given that none of the three is much use without the other two.

§3.1.6 — Why Chemistry Was Not Fast Enough

Cells had been signaling to one another for billions of years before animals existed, and animal bodies still use chemical signaling constantly. A hormone is released into the bloodstream by one tissue, travels through the body, and changes what other tissues do. Chemical signaling is cheap, it reaches everywhere, and it does not require any special apparatus beyond the ability to make the molecule and respond to it.

Its limitation is that it moves by diffusion, and diffusion is slow in a way that gets rapidly worse with distance. A small molecule crosses a micrometer, roughly the width of a bacterium, in about a millisecond. It crosses a millimeter in several minutes. It would cross a meter, the length of a large animal, in something on the order of a decade. The time does not grow in proportion to the distance; it grows with the square of the distance, so every doubling of distance quadruples the wait. Diffusion also spreads in every direction at once, which means a chemical signal cannot be aimed at one destination, and it lingers, which means it cannot easily be switched off. For a message like *grow* or *get hungry* or *begin reproducing*, none of that matters. For a message like *the left side of the body should contract, now*, all of it is fatal.

So it is worth being careful about what neurons actually contributed. They did not invent communication between cells, and they did not invent behavior; both were in place long before them. What they did was make a trade. Chemical signaling is cheap and reaches everything; neural signaling gives up the cheap broadcast reach and buys speed, distance, and aim in return. The two coexist in every animal body, doing the jobs each is suited to, and animals still run on hormones for everything that can afford to be slow.

A **neuron**, then, is the cell specialized to make that trade. It takes an event, whether in the world or elsewhere in the body, and converts it into a signal that travels fast, travels far without weakening, and arrives at a specific destination rather than everywhere. That is the computational-level answer to what a neuron is for, and it is why natural selection built neurons into the animal lineage and not into the other two: the lineages whose sense → decide → act loop ran faster were the lineages that ate rather than being eaten.

The answer comes with a bill attached, and the bill is the other thread of this chapter. Speed of that kind is not free, and the surplus that made animals possible in the first place is the same surplus that neurons spend in order to signal. A chapter that opens with life becoming rich enough to afford neurons will close, in §3.3, with the most expensive organ in the body.

We now have the problem and we know for whom it was urgent. The next question is not why but how. Granting that a neuron's job is to take signals in, combine them, and pass one along, what exactly is it computing? That is the algorithmic level.